PRANAY LACHULURI

pranayla@buffalo.edu | Buffalo, NY | LinkedIn | GitHub | Portfolio

EDUCATION

University at Buffalo, SUNY, MS in Computer Science (AI/ML) | GPA 3.7/4.0 — Aug 2025 – Jan 2027 (expected)
Coursework: Deep Learning, Intro to Machine Learning, Computer Vision & Image Processing, Statistical Data Mining II, Data-Intensive Computing, Computer Security, Operating Systems, Analysis of Algorithms, Data Models & Query Languages

Neil Gogte Institute of Technology, BE in Computer Science (AI/ML) | CGPA 7.1/10 — 2020 – 2024 | Hyderabad, India

EXPERIENCE

Graduate Research Assistant, University at Buffalo, ASAP (Autonomous Plastic-Sort) — Sep 2026 – Present | Buffalo, NY

- Collect and curate multi-sensor data for an ML-based material-identification system, operating hyperspectral, RGB and thermal cameras and mmWave radar on ROS 2.
- Calibrate the sensor rig (hyperspectral reflectance correction, multi-camera checkerboard calibration, radar range calibration) and record synchronized datasets under controlled conditions.
- Annotate images in CVAT and manage labeled datasets in FiftyOne, producing the training data for the project's detection and classification models.

Apprentice, Google/YouTube Client Operations, Genpact — Mar 2025 – Aug 2025 | India

- Performed data validation and quality-focused operations for a Google/YouTube client under strict client-defined policies and quality standards.
- Trained in generative AI, agentic AI and AI-assisted, prompt-driven workflows.

Data Science Intern, YHills — Apr 2023 – Jul 2023 | Remote

- Built and compared 7 scikit-learn and XGBoost models per task: fraud detection at **99.2%** accuracy (XGBoost), H1N1 vaccination prediction at **82.6%**, and taxi-fare regression at **R^2 0.76** (Random Forest).

PROJECTS

ShadowFix AI: Shadow Removal [Code] | PyTorch, U-Net, OpenCV, Gradio — 2026

- Built a 7.76M-parameter U-Net and compared RGB, residual and mask-guided variants on ISTD (1,197 train / 133 val / 540 test images) with AMP training and PSNR/SSIM plus shadow-region metrics.
- Best mask-guided model reached **30.19 dB PSNR** and **0.951 SSIM**, **+9.63 dB** over the raw input; shipped automatic and paint-the-shadow Gradio modes.

Trash Detection with YOLOv8 [Code] | YOLOv8, PyTorch, OpenCV, Gradio — 2026

- Fine-tuned YOLOv8m on a 59-class TACO-derived dataset at 832 px with AdamW, cosine LR, Mosaic and MixUp; 0.183 mAP@50 on 299 validation images (962 objects), +44% mAP@50-95 over a YOLOv8n baseline.
- Added confidence-weighted trash heatmaps, a 0–100 cleanliness score and an interactive Gradio assistant.

AI Security: Adversarial and LLM Attacks (coursework) [Code] | PyTorch, Hugging Face Transformers — 2025

- Ran FGSM/PGD attacks on a ResNet-50 classifier and jailbreaks on StableLM-3B; a hidden-HTML prompt injection hijacked Qwen2.5 in 60% of runs, and content filtering, system-prompt hardening and output validation each blocked it.

Transformers from Scratch: ViT and Character-level GPT (coursework) [Code] | PyTorch — 2026

- Implemented patch embedding, multi-head and causal self-attention and positional embeddings from scratch; ViT ablations on CIFAR-10 showed 2×2 patches beat 8×8 by 8.9 points and removing positional encoding cost 3 points.

Real-time Sign Language Detection [Code] | TensorFlow/Keras, LSTM, MediaPipe, OpenCV — 2024

- Recognizes 22 signs live from a webcam: MediaPipe extracts 21 hand landmarks per frame and a stacked LSTM classifies 30-frame sequences from a self-collected dataset, with a 10-frame agreement filter for stable labels.

Car Price Prediction App [Code] | scikit-learn, Flask — 2023

- Served a one-hot-encoded linear regression pipeline through a Flask app with dynamic inputs and input validation; reproducible training script with cross-validated R^2 of about 0.64.

SKILLS

Languages: Python, SQL, C, Java, JavaScript

ML / Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, CNNs, U-Net, YOLO, transfer learning, Hugging Face Transformers, Vision Transformers

Computer Vision & Sensing: OpenCV, MediaPipe, object detection, image restoration, hyperspectral and thermal imaging, mmWave radar, camera calibration

Data & Tools: PySpark, pandas, NumPy, MySQL, Git, Linux, Docker, ROS 2, Flask, Gradio, CVAT, FiftyOne, Jupyter, Colab